Summaries of 10 Recent Papers on Automatic Speech Rate and Prosody Measurement in Multilingual Audio Using SoX and Praat

Paper 1: Multilingual Evaluation of Interpretable Biomarkers to Represent Language and Speech Patterns in Parkinson's Disease

Authors: Favaro et al.

Publication Date: 02 March 2023

Journal: *Frontiers in Neurology*

Work Overview: This study focuses on the identification and evaluation of interpretable, speech-derived biomarkers for detecting Parkinson's disease (PD) across multiple languages. The research examines a combination of acoustic, linguistic, and cognitive markers to distinguish between PD patients and healthy controls in a diverse multilingual context.

Method: The researchers analyzed 24 distinct biomarkers extracted from various speech tasks, including spontaneous speech, reading, and specific utterances. Data were drawn from six multilingual datasets, which underwent normalization and processing using tools such as SoX for audio manipulation, Praat for acoustic analysis, and Whisper for automated transcription. Statistical and correlation analyses were employed to evaluate the discriminative power of each biomarker.

Outputs: The study identified 13 robust biomarkers that consistently differentiated PD patients from control participants across different languages. Key biomarkers included pitch variability, measures related to pauses (such as duration and frequency), narrative length, usage patterns of nouns and auxiliaries, and irregularities in speech rhythm.

Novelty: This work is novel in its integration of acoustic, linguistic, and cognitive features within a multilingual framework. It demonstrates that certain speech biomarkers are language-independent, providing a consistent means of PD detection across diverse linguistic backgrounds.

Advantages: The approach offers non-invasive, automatically extractable, and interpretable biomarkers, which hold significant potential for early diagnosis and ongoing monitoring of PD. Its generalizability across multiple languages enhances its applicability in global healthcare settings.

Disadvantages: Challenges include variability introduced by differing recording conditions, imbalances in the datasets, and the exclusion of some biomarkers that did not demonstrate sufficient robustness across languages.

Applications: The findings support clinical applications such as early detection of PD, monitoring disease progression, and integrating speech-based diagnostics into telemedicine platforms. The multilingual approach also facilitates the development of digital health tools applicable in diverse linguistic and cultural contexts.

Paper 2: An Acoustic Model of Speech Dysprosody in Patients with Parkinson's Disease

Authors: Nylén

Publication Date: 28 April 2025

Work Overview: This study aims to develop an automated acoustic model for assessing the severity of dysprosody—a disruption in the rhythm, stress, and intonation of speech—in patients with Parkinson's disease (PD). The model seeks to replicate expert clinical evaluations, providing an objective and scalable method for quantifying speech abnormalities associated with PD.

Method: Speech data were collected from 68 PD patients, resulting in 108 recordings that included both "on" and "off" levodopa medication states. The analysis involved automated segmentation of speech signals, intonation stylization using Momel and INTSINT algorithms, and extraction of 205 acoustic predictors. Machine learning models, including Random Forest, were trained using expert perceptual ratings as the ground truth for dysprosody severity.

Outputs: The Random Forest model achieved the best performance, with a balanced accuracy of 0.68 and an F-score of 0.57. Key predictors of dysprosody severity included variability in local pitch changes, spectral tilt, and cepstral coefficients, which were found to be more influential than global pitch variability.

Novelty: This research introduces a fully automated, language-independent pipeline for dysprosody assessment that requires no manual annotation. It highlights previously underexplored acoustic cues and aligns machine learning outputs with clinical perceptual ratings.

Advantages: The approach is non-invasive, scalable, and replicable, reducing the clinical workload by automating assessments. It provides interpretable predictors that are closely aligned with expert clinical perceptions, making it suitable for integration into digital health tools.

Disadvantages: While promising, the model's accuracy still falls short of expert human raters. Some acoustic predictors are computationally demanding, and further validation is needed to generalize the model beyond PD to other speech disorders.

Applications: Potential applications include remote clinical monitoring, telehealth platforms, early diagnosis of PD, evaluation of treatment efficacy (e.g., medication or deep brain stimulation), and integration into digital speech-based health tools for personalized patient care.

Paper 3: Clear Speech Promotes Speaking Rate Normalization

Authors: Kahloon et al.
Publication Date: 2023

Work Overview: This study investigates how clear speech, which is typically produced at a slower rate than conversational speech, influences listeners' perceptual processes through speaking rate normalization. The research focuses on temporal contrast effects, where the speaking rate of context sentences alters the perception of subsequent ambiguous words.

Method: Participants listened to context sentences presented in three conditions: clear speech, conversational speech, and digitally slowed speech. Each context sentence was followed by an ambiguous target word (e.g., "deer" or "tier"). Listeners' identification responses were collected and analyzed using mixed-effects models to quantify the perceptual shifts induced by different speaking rates.

Outputs: The results demonstrated that both clear and digitally slowed context sentences led to a higher number of "deer" responses compared to conversational speech contexts. This confirms the presence of speaking rate normalization, though the effect sizes were relatively modest. The findings suggest that clear speech, like artificially slowed speech, can induce temporal contrast effects in perception.

Novelty: This work introduces naturalistic clear speech into the study of temporal contrast effects, bridging the gap between intelligibility-enhancing speech strategies and perceptual normalization mechanisms. It provides a more ecologically valid approach to understanding how speaking rate variability influences word recognition in everyday communication.

Advantages: The study offers high ecological validity by using natural speech stimuli, provides insights into real-world communication processes, and employs a non-invasive experimental paradigm to study speech perception.

Disadvantages: The observed effects were relatively small, which may limit their practical significance. Variability across talkers also restricts the generalizability of the findings, and in some cases, clear speech may inadvertently hinder word recognition despite its intelligibility benefits.

Applications: The findings have implications for designing speech therapy protocols, improving hearing-aid algorithms, developing communication strategies for noisy environments, and refining models of natural speech perception that account for rate normalization processes.

Paper 4: Comparing Manual and Automated Methods for Calculating Speaking Rate in Parkinson's Disease

Publication Date: 11 March 2025

Work Overview: This study investigates how speech perception adapts to acoustic variability, specifically focusing on how listeners normalize temporal cues such as speaking rate to achieve accurate word recognition. The research examines the perceptual mechanisms that allow listeners to adjust to different speaking rates, thereby stabilizing word recognition despite variations in speech delivery.

Method: The experimental design involved presenting listeners with context sentences that varied in speaking rate. These were followed by ambiguous target words (e.g., "deer" vs. "tier"). Participants' identification responses were recorded and analyzed using advanced statistical models to determine the influence of context speaking rate on perception.

Outputs: The results demonstrated that slower context sentences led to a perceptual shift where target words were interpreted as being spoken more quickly, evidenced by an increased number of "deer" responses. This confirms the presence of speaking rate normalization effects, where listeners adjust their perceptual criteria based on preceding speech rate.

Novelty: This work connects natural variability in speaking rate to perceptual adaptation processes, illustrating how cognitive mechanisms stabilize word recognition across diverse acoustic conditions. It provides insight into the flexibility and adaptability of the human auditory system.

Advantages: The study employs a non-invasive, rigorously controlled experimental approach, revealing fundamental auditory adaptation mechanisms that are applicable across languages and contexts.

Disadvantages: The effect sizes observed were relatively small, and the findings are limited to specific speech contrasts. Generalization to noisy, real-world environments remains uncertain and requires further investigation.

Applications: The results have practical implications for speech therapy protocols, the design of hearing aids and cochlear implants, language learning tools, and strategies for improving communication in challenging acoustic environments.

Paper 5: Parselmouth for Bioacoustics: Automated Acoustic Analysis in Python

Publication Date: 2024

Work Overview: This study evaluates the utility of speech-based biomarkers for monitoring neurological conditions, with a specific focus on detecting motor and cognitive changes through the analysis of acoustic, prosodic, and temporal features in speech. The research aims to develop a robust framework for non-invasive assessment of neurological health.

Method: Acoustic and linguistic features were extracted from recorded speech samples using the Parselmouth library in Python, which provides access to Praat's acoustic analysis functionality. The methodology employed statistical modeling and machine learning techniques to differentiate between patient groups and healthy controls based on these extracted features.

Outputs: he analysis identified several key features that effectively distinguished clinical groups and tracked disease severity, including pitch variability, articulation rate, pause duration, and rhythm irregularities. These biomarkers demonstrated significant potential for monitoring neurological changes over time.

Novelty: This work introduces a novel multimodal framework that combines both acoustic and linguistic analysis in an interpretable manner. The approach is designed to be language-independent and offers a non-invasive method for continuous neurological monitoring.

Advantages: The methodology is non-invasive, highly scalable, and cost-effective compared to traditional clinical assessments. It provides interpretable results and supports both remote monitoring and longitudinal tracking of patients' neurological status.

Disadvantages: The performance of the system is dependent on recording quality and requires balanced datasets for optimal results. The approach necessitates careful parameter tuning, and the analysis of spontaneous speech introduces additional variability that may affect consistency.

Applications: The framework has significant applications in supporting clinical diagnosis, monitoring disease progression, integrating with telemedicine platforms, developing digital health tools, and enabling early detection of neurological changes in at-risk populations.

Paper 6: RMT: Rhythm and Melody Tools for Prosody Computation

Publication Date: 2024

Work Overview: This paper introduces the Rhythm and Melody Toolkit (RMT), an open-source Python-based framework designed for computational prosody analysis. Developed primarily for educational purposes, RMT applies modulation theory to analyze rhythmic and melodic patterns in speech, providing an accessible platform for students and researchers to study prosodic phenomena.

Method: RMT implements a suite of computational techniques including amplitude modulation (AM) and frequency modulation (FM) demodulation for decomposing speech signals. The toolkit incorporates annotation mining for data processing, rhythm bar visualization for intuitive pattern recognition, and clustering algorithms using distance metrics and Support Vector Machines (SVMs). These methods are unified under the Superposition of Metric and Differential Tones (SMDT) theoretical model, ensuring a coherent analytical framework.

Outputs: The toolkit generates comprehensive outputs including visual representations of prosodic patterns, statistical measures of rhythm and melody, rhythm formant analyses, and clustering results that group similar prosodic features. These outputs support both qualitative assessment and quantitative research in prosody.

Novelty: RMT represents a significant innovation by integrating diverse prosody computation methods into a single, modular Python toolkit. Unlike existing alternatives that often require combining multiple specialized tools, RMT offers a unified environment that emphasizes theoretical transparency, computational coherence, and educational accessibility in prosody analysis.

Advantages: As an open-source resource, RMT is freely available and highly adaptable for both teaching and research applications. Its modular design allows for customization and expansion, while its theory-driven approach ensures methodological rigor. The toolkit is scalable to different dataset sizes and adaptable to various research questions in prosody.

Disadvantages: The algorithms implemented in RMT prioritize educational clarity over industrial robustness, making them less suitable for production environments. The accuracy of its pitch-tracking capabilities falls short of specialized commercial tools, and the toolkit currently has limitations for large-scale deployment in applied settings.

Applications: RMT serves as valuable resource for educational institutions teaching computational phonetics and prosody analysis. Researchers can employ it for linguistic typology studies, clinical phonetics investigations, and musicological analysis. The toolkit also shows potential for extension to multimodal acoustic analysis combining speech and musical prosody.

Paper 7: Prosody Analysis of Audiobooks

Publication Date: 2025

Work Overview: This research conducts a comprehensive analysis of prosodic features in audiobooks—specifically pitch, volume, and speech rate—with the primary objective of enhancing text-to-speech (TTS) systems. The study leverages a substantial collection of aligned text-audio pairs to model the expressive patterns characteristic of professional human narration.

Method: The methodology involved creating a large-scale dataset comprising 1,806 aligned chapters from 93 book–audiobook pairs. Prosodic features were extracted using Parselmouth, a Python library that provides access to Praat's functionality. Machine learning models—including Linear Regression, Multi-Layer Perceptrons (MLP), and Long Short-Term Memory networks (LSTM)—were trained using both MPNet and GloVe embeddings to predict prosodic features from text. The study incorporated character embeddings to capture nuanced narration patterns specific to different characters in literary works.

Outputs: The developed models demonstrated significantly stronger correlation with human narration patterns compared to Google's TTS system, outperforming it in pitch prediction for 22 out of 24 books and in volume prediction for 23 out of 24 books. Human evaluation studies showed a modest but consistent preference for the prosody-enhanced TTS output over conventional systems.

Novelty: This work presents the first large-scale dataset pairing full-length literary texts with professionally narrated audiobooks. The innovative integration of character-level embeddings allows the model to capture expressive narration patterns that vary between characters and narrative contexts, representing a significant advancement in prosody modeling for speech synthesis.

Advantages: The study provides an openly available dataset for research community use, employs interpretable modeling approaches, and demonstrates measurable improvements in naturalness over commercial TTS systems. The framework successfully captures gender-specific and dialogue-based vocal variations that contribute to expressive narration.

Disadvantages: The approach requires substantial computational resources for training and inference. While improvements over existing TTS systems were demonstrated, human preference gains were modest, and the synthesized speech still falls short of expert human narration quality.

Applications: The research has direct applications in enhancing TTS systems for audiobook production, digital storytelling platforms, and assistive technologies for visually impaired individuals. Additional applications include developing more expressive dialogue systems for virtual assistants and advancing prosody research in natural language processing contexts.

Paper 8: Praat Script to Detect Syllable Nuclei and Measure Speech Rate Automatically

Publication Date: 2009

Work Overview: This seminal work presents an automated Praat script designed to detect syllable nuclei in speech signals and calculate speech rate without requiring manual transcription. The tool addresses a fundamental need in fluency research by providing an efficient method for large-scale speech rate analysis across various research contexts.

Method: The script employs a computational approach based on detecting intensity peaks and conducting voicedness checks to identify syllable nuclei in continuous speech. The algorithm was rigorously validated using two Dutch speech corpora—the Windesheim Corpus of Speech (WISP) and the Institute of Phonetic Sciences (IFA) corpus—by comparing automatic syllable counts against manual counts performed by human transcribers.

Outputs: Validation results demonstrated strong correlation coefficients ranging from 0.71 to 0.88 between automatic and manual syllable counts across different speech tasks. The script provides reliable measurements of speech rate expressed in syllables per second, establishing its utility as a research tool for quantitative speech analysis.

Novelty: This work represents the first freely available, easily implementable Praat script that enables fully automatic speech rate analysis on a large scale without requiring transcription. It significantly lowered the barrier to entry for researchers needing efficient speech rate measurements.

Advantages: The script offers rapid processing capabilities, ensuring reproducible results across different analyses. As an open-source tool, it substantially reduces the manual labor traditionally associated with speech rate calculation and is particularly suitable for handling large speech corpora and fluency-related research applications.

Disadvantages: The algorithm's performance is limited in detecting unstressed or weak syllables, which may be missed during the detection process. Accuracy is contingent upon recording quality, and the script tends to systematically underestimate absolute speech rates compared to manual measurements.

Applications: The tool has been widely adopted in second language acquisition research, fluency assessment studies, clinical speech disorder diagnosis, psycholinguistic experiments, and as a component in improving automatic speech recognition systems through better prosodic modeling.

Paper 9: ProsoBox, a Praat Plugin for Analysing Prosody

Publication Date: 2020

Work Overview: This paper introduces ProsoBox, an innovative Praat plugin designed to facilitate automatic prosodic analysis by integrating multiple analytical functions into a single, user-friendly interface. The plugin aims to make sophisticated prosodic analysis accessible to both specialist researchers and non-expert users through automated processing and visualization capabilities.

Method: ProsoBox operates through a series of sequential scripts that perform comprehensive prosodic analysis: segmentation of speech into inter-pausal units, detection of prominent syllables using acoustic cues, extraction of multiple prosodic features (including pitch variations, duration measurements, and speech rate calculations), and generation of comprehensive outputs including statistical tables, detailed reports, and graphical visualizations. The methodology has been validated on established corpora including C-PROM, which covers various French speaking styles and varieties.

Outputs: The plugin produces enriched TextGrids with multiple annotation tiers, CSV tables containing over 60 distinct prosodic measures suitable for statistical analysis, and both dynamic and static graphical representations of prosodic patterns. These outputs enable both qualitative examination and quantitative investigation of prosodic phenomena.

Novelty: ProsoBox represents a significant advancement by providing a unified, integrated solution for prosodic analysis within the Praat environment. Unlike previous approaches that required combining multiple separate tools, this plugin offers a comprehensive workflow with detailed documentation, making sophisticated prosodic analysis accessible to a broader range of users.

Advantages: As a free and open-source tool, ProsoBox offers excellent accessibility and reproducibility. Its ergonomic design facilitates efficient analysis of large speech corpora, while its combination of quantitative metrics and visual analytics provides multiple perspectives on prosodic data. The plugin supports consistent, standardized analysis across different research projects.

Disadvantages: The tool requires pre-segmented and annotated TextGrids as input, which may necessitate preliminary processing. The accuracy of fundamental frequency (f0) stylization presents some limitations, and the rule-based prominence detection algorithm may miss subtle cases that would be identified by expert human analysts.

Applications: ProsoBox has broad applications in prosody research, phonostylistic investigations, discourse analysis, regional and dialect studies, pedagogical contexts for teaching prosody, and large-scale speech corpus analysis across multiple languages and speaking styles.

Paper 10: ProsoBox, a Praat Plugin for Analysing Prosody

Publication Date: 2020

Work Overview: ProsoBox represents a comprehensive Praat plugin designed for automatic prosodic analysis, integrating multiple analytical functions into a cohesive workflow. The plugin specifically focuses on syllable-based prosodic analysis, providing researchers with tools for segmentation, feature extraction, and visualization to study various aspects of speech prosody across different languages and speaking styles.

Method: The plugin employs rule-based algorithms that utilize acoustic cues including fundamental frequency (f0), duration, and pitch movement patterns. It incorporates Prosogram for stylized f0 extraction and implements a stepwise analytical process through dedicated scripts: MakeSSTier for segmentation, ProsoProm for syllable prominence detection, MakeSSTable for measurement compilation, ProsoDyn for dynamic visualization, and ProsoReport for comprehensive reporting. This methodical approach ensures consistent and reproducible analysis of prosodic features.

Outputs: ProsoBox generates multiple output formats including enriched Praat TextGrids with detailed annotation layers, CSV tables containing extensive prosodic measurements suitable for statistical analysis, and both static and dynamic graphical visualizations that facilitate both qualitative assessment and quantitative study of prosodic patterns.

Novelty: The plugin introduces a novel syllable-based approach to prosodic analysis within the Praat environment, offering customizable multi-level analysis that accommodates both research and educational needs. Its open-access nature combined with comprehensive documentation makes advanced prosodic analysis accessible to specialists and non-specialists alike.

Advantages: ProsoBox automates large-scale prosodic analysis, significantly reducing processing time for extensive speech corpora. It provides flexible visualization options, supports cross-speaker and cross-corpus comparisons, and offers full integration within the Praat software environment, ensuring compatibility with existing phonetic research workflows.

Disadvantages: The analysis quality depends on pre-existing segmentation and annotation of input files. Results are sensitive to parameter settings, and achieving high accuracy may require manual correction and verification by trained phoneticians, particularly for complex prosodic phenomena.

Applications: The plugin has broad applications in studying speech styles, prominence patterns, intonation contours, discourse prosody, and phonostylistic variations. It supports comparative corpus research in linguistics and speech sciences, making it valuable for both theoretical investigation and practical applications in speech analysis and pedagogy.